

Syllabus

Discrete Mathematics

Teacher Information

- Email: dt323259@gmail.com
- Phone: 435-619-1784

Books

- Mathematics A Discrete Introduction 3rd Ed. by Scheinerman (Provided by Teacher)

Grading

- Homework 50% (Homework is due the week after it is assigned)
- Tests 30%
- Midterm/Final 20% (Also replaces your lowest test score for the semester)
- Extra Credit 5%

Schedule

- Week 1
 - Propositional & Predicate Logic
 - Homework: Handout
- Week 2
 - Sections 1-4 (Logic, Writing Mathematics, Definitions, Theorems)
 - Homework 1 : 1.1, 3.1, 3.3, 3.5, 3.6, 3.7, 3.9, 3.11, 3.13 part a), 4.1-4, 4.7
 - Extra Credit : 4.12 (Required to read the problem for class discussion)
- Week 3
 - Sections 5-7 (Proofs, Counterexample, Boolean Algebra)
 - Homework 2 : 5.1 - 13, 6.1, 6.3, 6.5, 6.11, 7.1-7, 7.11, 7.13
 - Extra Credit : 5.18, 5.22

- Week 4
 - Chapter 1 Test
 - Sections 8 (Lists: Ordered Collections)
 - Homework 3 : 8.1, 8.3, 8.7, 8.12, 8.13
 - Extra Credit : 8.19
- Week 5
 - Sections 9-11 (Factorial, Sets & Subsets, Quantifiers)
 - Homework 4 : 9.1, 9.2, 9.4, 9.5, 9.9, 9.10, 10.1 - 10, 11.1 - 7
 - Extra Credit : 9.11 or prove 9.9
- Week 6
 - Sections 12-13 (Set Operations, Combinatorial Proofs)
 - Homework 5 : 12.1-3, 12.5, 12.7, 12.11, 12.21, 12.22, 13.2, 13.4, 13.5, 13.7
 - Extra Credit : 12.29
- Week 7
 - Chapter 2 Test
 - Sections 14 (Relations: Sets of Ordered Pairs)
 - Homework 6 : 14.1 - 8
 - Extra Credit : 14.15
- Week 8
 - Sections 15-16 (Equivalence Relations, Partitions)
 - Homework 7 : 15.1 - 7, 16.1 - 5, 16.7, 16.15
 - Extra Credit : 15.12
- Week 9
 - Sections 17 (Binomial Coefficients)
 - Homework 8 : 17.1 - 6, 17.11, 17.14
 - Extra Credit : 17.19
- Week 10
 - Chapter 3 Test
 - Sections 20 (Proof by Contradiction)
 - Homework 9 : 20.1, 20.2, 20.3, 20.5, 20.6, 20.7
 - Extra Credit : 20.10
- Week 11
 - Sections 21-22 (Proof by Smallest Counterexample, Proof by Induction)
 - Homework 10 : 21.2, 21.3, 21.4, 21.8, 21.11, 22.4 a-d, 22.5 (not c)
 - Extra Credit : 22.11

- Week 12
 - Chapter 4 Test
 - Midterm Review
- Week 13
 - Midterm
- Week 14
 - Sections 24 (Functions: Set Theoretical Definition)
 - Homework 11 : 24.1, 24.3, 24.5
 - Extra Credit : 24.16
- Week 15
 - Sections 26-27 (Composition, Permutations)
 - Homework 12 : 26.1, 26.5, 26.6, 27.2, 27.5, 27.6
 - Extra Credit : 26.9
- Week 16
 - Chapter 5 Test
 - Sections 30 (Probability: Sample Spaces)
 - Homework 13 : 30.1, 30.2, 30.3, 30.4, 30.5
 - Extra Credit : 30.10
- Week 17
 - Sections 31-32 (Exents, Conditional Probability & Independence)
 - Homework 14 : 31.1, 31.3, 31.4, 31.7, 32.1, 32.4
 - Extra Credit : 32.31
- Week 18
 - Sections 33-34 (Random Variables, Expected Value)
 - Homework 15 : 33.1, 33.2, 34.1, 34.2, 34.4
 - Extra Credit : 33.9
- Week 19
 - Chapter 6 Test
 - Sections 35 (Number Theory: Dividing)
 - Homework 16 : 35.1, 35.2, 35.8
 - Extra Credit : 35.12
- Week 20
 - Sections 36-37 (Greatest Common Divisor, Modular Arithmetic)
 - Homework 17 : 36.1, 36.2, 37.1, 37.2, 37.3, 37.4
 - Extra Credit : 36.11

- Week 21
 - Sections 38-39 (Chinese Remainder Theorem, Factoring)
 - Homework 18 : 38.1, 38.2, 38.3, 39.2, 39.5, 39.6
 - Extra Credit : 39.16
- Week 22
 - Chapter 7 Test
 - Sections 40 (Group Theory)
 - Homework 19 : 40.1, 40.2, 40.4, 40.7
 - Extra Credit : 40.14
- Week 23
 - Sections 41-42 (Group Isomorphisms, Subgroups)
 - Homework 20 : 41.1, 41.2, 41.3, 42.1, 42.2
 - Extra Credit : 41.4
- Week 24
 - Sections 43 (Fermat's Little Theorem)
 - Homework 21 : 43.1, 43.2, 43.3, 43.4, 43.5
 - Extra Credit : 43.8
- Week 25
 - Chapter 8 Test
 - Final Review
- Week 26
 - Final
 - Nothing after this will affect your grade
- Week 27 (Cushion Week)
 - Section 44-45 (Public Key Cryptography, Rabin's Method)
- Week 28 (Cushion Week)
 - Section 46 (Public Key Cryptography: RSA Encryption)